

7. (a) In dry conditions, a Formula One (F1) car can accelerate from 0 to 30 m/s in 1.5 seconds in a straight line.

(i) State the change in velocity. [1]

Change in velocity = m/s

(ii) Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

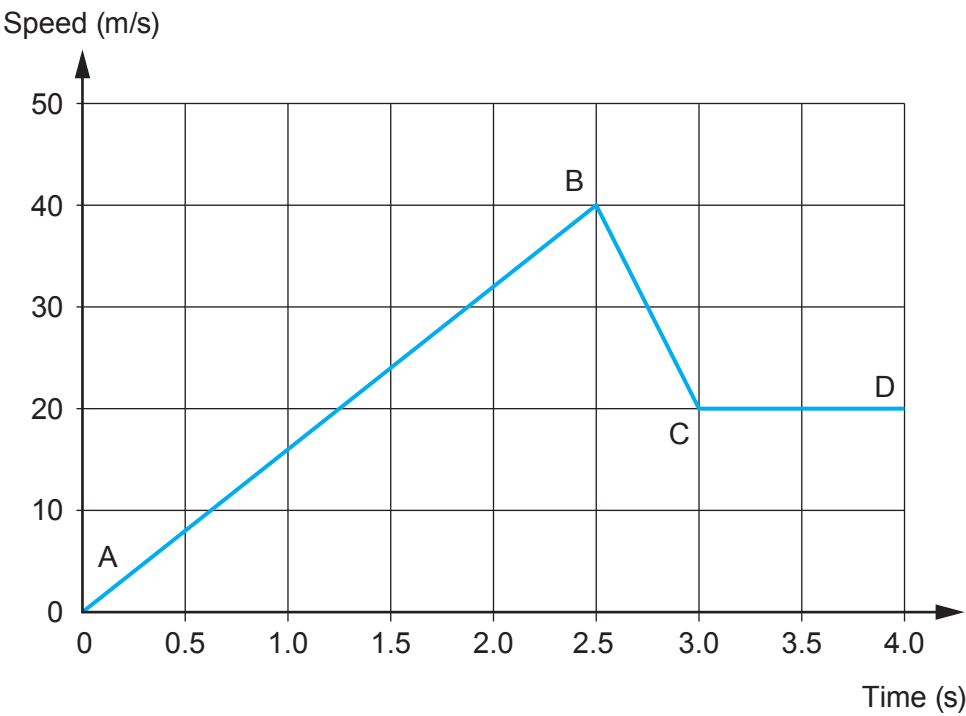
to calculate the acceleration of the racing car. [2]

Acceleration = m/s²

(b) The photograph shows F1 cars lined up on the grid.



The graph below shows how the speed of a F1 racing car changes at the start of a race as it leaves the grid and goes around the first bend.



- (i) **Use data from the graph** to describe the motion of the car during the time shown.
No calculations are required. [6 QER]

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- (ii) The car travels 85 m during the time shown by the graph.
Use an equation from page 2 to calculate the mean speed of the car during this time. [3]

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Mean speed = m/s

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